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Tutorials in Clinical Research: VII. Understanding Comparative Statistics (Contrast)—Part B: Application of T-Test, Mann-Whitney U, and Chi-Square

J. Gail Neely, MD; James M. Hartman, MD; James W. Forsen, Jr., MD; Mark S. Wallace, MD

Objective: **This tutorial on comparative statistics has been written in two complementary segments. The first paper (part A) focused on explaining the general concepts of the null hypothesis and statistical significance. This second article (part B) addresses the application of three specific statistical tests. These two articles should be read sequentially and the first article should be available for reference while one reads the second.** ***Study Design:*** **Tutorial.** ***Methods:*** **The authors met weekly for 10 months to discuss clinical research articles and the applied statistics. The difficulty was not the material but the effort to make it easy to read and as short as possible.** ***Results:*** **The article discusses the application of three common statistical indexes of contrast, χ^2, Mann-Whitney *U*, and Student *t*-test and other concepts, such as sample size, degrees of freedom, errors, power, and confidence intervals.** ***Conclusions:*** **Statistical tests generate a number known as a *statistic* (χ^2, *U*, *t*), which is sometimes called a "critical ratio" because it helps us to make a decision. This number is then associated with a probability, or *P* value. Sample size is a crucial element in the initial design of a research project and in the subsequent ability of the results to show statistical significance if the difference is clinically important. The example data used in this paper demonstrate the application of the three specific tests and illustrate the effect of sample size on the results.** ***Key words:*** **research/methods, research design, clinical protocols, statistics/methods.**

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This tutorial on comparative statistics has been written in two complementary segments to reduce the size and increase the comprehension of each. The first paper focused on explaining the general concepts of the null hypothesis and statistical significance as they apply to statistical indexes of contrast used in comparisons between two groups. This second article addresses the application of three specific statistical tests. It is advisable to read these two articles sequentially and to have the first article at hand for reference while reading the second.

Three classic tests that commonly serve as statistical indexes of contrast are reviewed.[1] These are the Student *t*-test, the Mann-Whitney *U* test, and the χ^2 test. The Student *t*-test is characteristically used for comparing continuous variables (such as age, weight, blood loss, and other equal interval dimensional data) between two groups. The Mann-Whitney *U* test is often used to compare ordinal data (such as grades of pain, depths of anesthesia, and degrees of facial paralysis) between two groups. The χ^2 test is classically used to evaluate dichotomous data (such as success/failure, live/dead, positive/negative), although it can also be used for categorical variables with three or more categories (Table I).

The value of being conversant with these few tests was emphasized by a study by Emerson and Colditz.[2] They reviewed 760 articles in *The New England Journal of Medicine* and determined that a working knowledge of *t*-tests, χ^2 tests, and nonparametric tests such as the Mann-Whitney *U* test alone would give the reader "complete statistical access" to 75% of the articles.*

At the end of each subsection, a summary is provided to assist the reader in a rapid overview of the article. Formulae and calculations generally have been omitted

From the Clinical Research Working Group, Department of Otolaryngology–Head and Neck Surgery, Washington University School of Medicine, St. Louis, Missouri, U.S.A.

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Correspondence to J. Gail Neely, MD, FACS, Professor and Director, Otology, Neurotology, Base of Skull Surgery and Director of Research, Department of Otolaryngology–Head and Neck Surgery, Washington University School of Medicine, 660 South Euclid Avenue, Box 8115, St. Louis, MO 63110, U.S.A. E-mail: jgneely@aol.com

*More recently, tests such as the Kaplan-Meier life tables and associated statistical tests, such as Cox regression models and multivariable analysis, particularly logistic regression, might need to be added to this list. However, this article will not consider these tests.

TABLE I.
Choice of Statistical Tests for Comparisons.

Dependent (Outcome) Variable	Independent Variable*	Test†
Continuous	Dichotomous (unpaired)	**Student *t*-test**
Continuous	Dichotomous (paired)	Paired *t*-test
Continuous	Nominal	ANOVA
Ordinal	Dichotomous (unpaired)	**Mann-Whitney *U***
Ordinal	Dichotomous (paired)	Wilcoxon matched-pairs signed-ranks test
Ordinal	Nominal	Kruskal-Wallis test
Dichotomous	Dichotomous (unpaired)	**χ^2 test or Fisher exact probability test**
Dichotomous	Dichotomous (paired)	McNemar χ^2 test
Dichotomous	Nominal	χ^2 test
Nominal	Nominal	χ^2 test

*"Unpaired" means that the samples are not related; for example, treatment A versustreatment B in two different groups. "Paired" means that the samples are not independent;for example, before treatment and after treatment in the same group of subjects.
†The bolded tests are those described in the two-part tutorial on comparative statistics. Data from reference[7].

from the text but are provided in tables and footnotes for interested readers. Figures attempt to characterize the important basic concepts in easy to understand and remember graphic illustrations.

APPLICATION OF THE THREE TESTS

To provide a simple example of three types of data (dichotomous, ordinal, and continuous) suitable for analysis by the three statistical tests, a table of actual observations was constructed to illustrate nonrandom clinical data from our own practice describing the outcomes from three different assessment tools used on two groups of subjects (group A and group B) 1 month following resection of solitary vestibular schwannomas. For purposes of illustration, the reader should consider group A as if surgical dissection was conducted from medial to lateral along the facial nerve and group B as if surgical dissection was conducted from lateral to medial. To maintain focus on the topic at hand, further details are not provided. The complete data for these analyses are shown in Table II.

The answers to two fundamental questions are sought: (1) Are the outcomes of the two groups statistically significantly different and (2) how large should the sample size be for a randomized trial to obtain a more definitive answer?

χ^2 Test

For the dichotomous outcome variable (normal vs. abnormal), the standard unpaired (Pearson) χ^2 test was chosen. The next step was to construct a 2×2 contingency table (Table III) and insert the frequency counts. The rows were designated by groups (A and B) and the columns were designated by "normal" and "abnormal." There were four normal and six abnormal in group A, and three normal and seven abnormal in group B. The final step was to enter the 2×2 table into an easy-to-use statistical pro-

TABLE II.
Facial Function Results in Two Groups of Patients 1 MonthAfter Acoustic Tumor Removal.

GROUP A				GROUP B			
Patient ID #	Dichotomous Outcome Variable*	Ordinal Outcome Variable†	Continuous Outcome Variable‡	Patient ID #	Dichotomous Outcome Variable*	Ordinal Outcome Variable†	Continuous Outcome Variable‡
1	Normal	I	0.94	1	Normal	I	0.94
2	Normal	I	0.98	2	Abnormal	VI	0.13
3	Abnormal	II	0.87	3	Abnormal	II	0.87
4	Normal	I	0.95	4	Normal	I	0.90
5	Abnormal	II	0.85	5	Abnormal	III	0.55
6	Abnormal	V	0.40	6	Abnormal	V	0.29
7	Abnormal	V	0.30	7	Abnormal	VI	0.04
8	Normal	I	0.99	8	Normal	I	0.90
9	Abnormal	III	0.55	9	Abnormal	III	0.55
10	Abnormal	III	0.52	10	Abnormal	III	0.52

*Dichotomous outcome variable = patient self-assessment
†Ordinal outcome variable =House-Brackmann facial grading scale[12]
‡Continuous outcome variable = Facial AnalysisComputerized Evaluation of Neely and Cheung (F.A.C.E.)[13]

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TABLE III.
2 × 2 Contingency Table.

	OUTCOME		
INTERVENTIONS:	Normal	Abnormal	
Group A	4	6	10
Group B	3	7	10
	7	13	20

gram and run a χ^2 test.[3,4] Alpha was set at 0.05. Because we did not know in advance if group A or group B would be better, a two-tailed arrangement was chosen for the test. The χ^2 statistic was 0.2198 with 1 degree of freedom. The *P* value associated with that statistic was 0.6392, which indicated that the difference between group A and group B was not statistically significantly different.

The computerized statistical program provided a warning that the expected values were less than 5 in more than 20% of the cells and suggested a calculation of the exact *P* value using the Fisher Exact Probability Test.[3] If the total number of values in the table is less than 20, or if any cell is less than 5, then the Fisher Exact Test is required.[5] The Fisher exact probability was $P = 1.00$, which was not statistically significant.

Mann-Whitney U *test*

For the ordinal outcome variable in Table 2, a nonparametric test, the Mann-Whitney *U* test, was chosen. The first step was to convert the Roman numerals to Arabic numbers for calculation. The next step was to paste the array of the ordinal values for group A into one column and the values for group B into an adjacent column in a statistical program and run the Mann-Whitney *U* test in a two-tailed arrangement. The *U* statistic was 39 and the associated *P* value was 0.4359. With the alpha set at 0.05, this is not statistically significant.

Student t-Test

For the continuous variable in Table 2, the Student *t*-test was chosen. The continuous data array for group A was pasted into the first column of a statistical program and the array for group B was pasted into the adjacent column and a Student (two group) *t*-test was run. The *t* statistic was 1.234 with 18 degrees of freedom; this was associated with a *P* value of 0.2329. The difference between the means was 0.1660. The 95% confidence interval (CI) about the mean of the difference was −0.4485 to 0.1165, which includes the value of 0, which is another way of showing it was not statistically significant.

These data indicate that the outcomes of the two groups are not statistically significantly different on any of the three ways the outcomes were described.

Sample Size

Perhaps these data were not statistically significant because the sample size was too small, which unfortunately is all too common. If a new experiment were to be conducted, just how big should these numbers be and what type of outcome measurement is most likely to show a difference? Preliminary data can help with this answer. Computer statistical programs easily calculate sample sizes; however, some data must be available to make these calculations. These kinds of data can be obtained from the literature, from clinical practice, as in this example, or, preferably, from a well-designed feasibility study, which is far more accurate.

For unpaired *t*-tests, the following data are needed to calculate sample size: (1) the expected difference between group means; (2) the expected standard deviation (SD) of each group, assuming that they are equal; (3) the statistical power that is desired; and (4) the alpha you choose. For proportions, the data needed are (1) the group 1 proportion (usually "success"); (2) the group 2 proportion (usually "success"); (3) the desired power; and (4) alpha.[3,6]

Using the observed differences in the small sample, calculations using the dichotomous data for proportions showed that approximately 350 subjects would be required in each group to determine statistical significance; thus, the total sample size of both groups would be 700 subjects. Calculations using the ordinal data showed that approximately 102 subjects would be required in each group; the total sample size of both groups would be 204 subjects. Calculations using the continuous data for a *t*-test showed that 50 subjects would be required for each group; the total sample size of both groups would be 100 subjects.†

To illustrate the effect of sample size and data type, the same values for facial function were copied and pasted twice again into the same spread sheet, giving groups of 30 (i.e., three times the original sample size). In this exercise, the χ^2 of the dichotomous data showed a *P* value of 0.4168 and a Fisher Exact Test *P* value of 0.5889. The Mann-Whitney *U* test of the ordinal data showed a *P* value of 0.1436. Neither of these is statistically significant, but the *P* values are smaller than in the group of 10. In both these cases, power assessments would be important to show just how much of an error would be made if the nonsignificant results were taken to be true. The continuous data showed a *P* value of 0.0306, which is statistically significant, and the power assessment would be moot.

Summary

These calculations would suggest that a continuous scale outcome measure might be a better way to show significant differences between the groups. The dichotomous and ordinal data were less robust because information in the continuous variable was lost when converting

†Calculations for the dichotomous data for proportions used: group 1 = 0.3; group 2 = 0.4; power = 0.80; alpha = 0.50. Calculations for the ordinal data, approaching the data as continuous, used: group A and group B, respectively, means 2.4 and 3.1; medians 2 and 3; and standard deviations 1.58 and 1.97, averaging 1.77. (Expected difference between means = 0.7; expected standard deviation = 1.77; power 0.80; alpha 0.50). If the ordinal data were rendered dichotomous by combining grades I and II and combining grades III and VI, the calculations used were: group 1 = 0.6; group 2 = 0.4; power = 0.80; alpha 0.05.) Calculations for the continuous data for a *t*-test used: group A and group B, respectively, means 0.7350 and 0.5690; standard deviations 0.26 and 0.33, averaging 0.30. (Expected difference between group means = 0.1660; expected standard deviation = 0.30; power = 0.80; alpha = 0.05.)

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to ordinal or dichotomous scales. Less information was lost in going to ordinal data, so that, as expected, the sample size required was not as large as for the dichotomous outcome data.

One of the major benefits of knowing about the different types of data and the ways to calculate sample sizes is in the speed and accuracy of the critical evaluation of the literature for practice. For example, if the sample size is small and the data are dichotomous, a large difference in proportions or a huge sample size would be required to demonstrate statistical significance.

DEGREES OF FREEDOM, SAMPLE SIZE, AND ERRORS

Degrees of Freedom

When determining the P value associated with the t-test and χ^2 statistic, the degrees of freedom (df) must be considered. *Degrees of freedom* refers to the "number of observations that are free to vary".[7] This concept is easy to understand by looking at the 2 × 2 table (Table III). The marginal values are calculated by the actual observed values located in the internal cells. Under the null hypothesis, one assumes that the marginal values are fixed and then determines how many of the four internal cells are free to vary. In a 2 × 2 table, the answer is one, because the rows and columns must add up to their respective totals. For contingency tables for the χ^2 test, the formula for degrees of freedom is df = (R − 1) (C − 1), in which R stands for rows and C stands for columns. For t-test comparison between the means of two groups of continuous data, the formula is df = $N_A + N_B - 2$, where N_A refers to the number of observations in treatment group A, N_B refers to the number of observations in treatment group B, and 2 refers to the number of times a mean is taken.[7]

The probability associated with the value of the statistic (χ^2 or t) is contingent on the degrees of freedom. When one calculates the statistic and looks up its associated P value from a table of tail probabilities, the degree of freedom is required; the same is true when the computer program does all this automatically.

For the Mann-Whitney U-test, the probabilities are contingent on the number of observations.

There are numerous statistical texts that go into this in detail; interested readers may wish to read some of the references listed at the end of this article.

Sample Size and Errors

Sample size refers to the number of subjects, or observations in, or required for, a single group (N); the total sample size (2N for two groups) refers to the total number in or required for a study.[5,7,8] Statistical programs calculate required sample sizes easily. Different methods may be used, however; therefore, some degree of understanding of what these programs are doing is useful (Table IV).

The primary issue under investigation is to determine whether a real, clinically important difference between treatment A and treatment B exists. This issue pivots on the statistical significance of the results, which hinges on whether the observed results are inconsistent enough with the null hypothesis to allow less than 5% probability of error if the null hypothesis is rejected (assuming we set alpha at 5%, 0.05). If this is the only concern, fewer subjects will be needed and the sample size can be calculated considering only the alpha error, also called a type I or false-positive error.

On the other hand, if the study results are not statistically significant, but the difference observed seems to be clinically important, we are in quite a bind, because we may be missing a true difference. Are these results by chance, or would they be statistically significant, and thus probably not by chance, if only we, or the authors, had studied more subjects? This is to say, are we making a beta error, also known as a type II or false negative error? (Table V and Fig. 1).

Using sample size calculations that consider both alpha and beta errors could keep us out of this dilemma, because it shows just how many additional subjects would be needed to minimize the probability of this beta error as well.[5] However, it does make the calculations more difficult. We do not want to have the cost of unnecessary research subjects, but neither do we want to dismiss a clinically important treatment just because we could not reach statistical significance from samples that were too

TABLE IV.
Calculating Sample Size (N) for Each Group.

General formula	$N = (Z_\alpha)^2(v)/d^2$
Paired t-test, alpha error only	$N = (Z_\alpha)^2(s)^2/(\delta)^2$
Student's t-test, alpha error only	$N = 2(Z_\alpha)^2(s)^2/(\delta)^2$
Student's t-test, alpha and beta errors	$N = 2(Z_\alpha + Z_\beta)^2(s)^2/(\delta)^2$
Tests of differences of proportions, alpha error only	$N = 2(Z_\alpha)^2(\bar{p})(1 - \bar{p})/(\delta)^2$
Tests of differences of proportions, alpha and beta errors	$N = 2(Z_\alpha + Z_\beta)^2(\bar{p})(1 - \bar{p})/(\delta)^2$

v = variance = $(s_p)^2$, which is the estimate of the combined, or pooled (s_p^2), variance ofthe comparison groups (estimated from pilot work or the literature). $s_p^2 = \Sigma[(X_A - \bar{X}_A)^2 + \Sigma(X_B - \bar{X}_B)^2]/(N_A + N_B - 2)$. X_A and X_B = individual values in group A and group B. $\bar{X}_A$ and $\bar{X}_B$ = means of group A and B. N_A is the number of observations in group A;N_B is the number of observations in group B; and 2 is the number of means taken. Z_α = Zscore for a value of alpha; when alpha set at 0.05 for two-tailed test, Z_α = 1.96. d =distinction = estimated difference between groups (δ). Z_β = Z score for a value of beta(β); when beta set at 0.2 (always one tailed), Z_β = 0.84. $(\bar{p})(1 - \bar{p})$ = estimate of thecombined variance of the groups of proportional data ($\bar{p}$ is the mean success in the twogroups combined)
Data from 1, 5, and 7.

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TABLE V.
Alpha and Beta Relationship with Null Hypothesis (H_0)(Probabilities of Occurrence in Parentheses).

Statistical Conclusion (reject or concede null hypothesis)	Truth	
	H_0 True	H_0 False
Reject	False-positive, type I error (α)	True-positive ($1 - \beta$)
Concede	True-negative ($1 - \alpha$)	False-negative, type II error (β)

Data from references 5 and 11.

small. Merely adding a few more cases after the analysis is not proper, because it biases the results.

These problems usually can be avoided by beginning with the very best estimate of the smallest clinical difference deemed important and using the sample size calculation for alpha errors only. This strategy tends to ensure that if statistically significant results are not obtained, the differences would not be clinically important, even if statistically significance could be found by a larger sample size.[5] To help explore what numbers might be candidates for the smallest clinically important difference, and at the same time might be obtainable with a study of sufficient power to ensure not rejecting treatments of value, Goodman and Berlin[9] developed a simple formula to calculate the smallest observed difference that is statistically significant at a given power of 0.80 or 0.90.‡ Note: using a small difference you wish to detect has a more profound effect on the sample size needed than any of the other variables. Usually it will take huge numbers to detect a very small difference.

This problem highlights the value of carefully done pilot work to have feasible data on which to do these calculations. Without pilot feasibility studies, the only other recourse is to try to determine the clinically meaningful difference and the estimated variances from practice or the literature; this may not be as satisfactory as desired. In reading an article, the reader should look for statements that indicate that the authors estimated the needed sample size a priori.

‡For a given power and clinically important difference, the predicted 95% confidence interval can be calculated by taking 0.7 times the difference for which there is an 80% power and adding and subtracting that value to the actually observed difference. If 90% power is desired, 0.6 is the multiplier. The 0.7 or 0.6 times the difference for which there is a desired power is the smallest observed difference that is statistically significant.

Summary

Degrees of freedom refers to the number of observations free to vary; this is crucial to determine the probability associated with a statistic. *Sample size*, after rigorous research design to control bias, is one of the most important elements to ensure that the study is capable of showing a clinically important difference if one really exists. An *alpha*, or *type I*, error exists if a false-positive error has occurred; this means that the null hypothesis has been rejected and a difference is said to exist when, in fact, there is no difference. A *beta*, or *type II*, error exists if a false-negative error has occurred; this means that the null hypothesis is conceded and no difference is said to exist when, in fact, a difference is really present, if only the sample size had been big enough to show it.

BETA AND POWER

Beta (β) defines the point at which we are comfortable in being wrong if we concede the null hypothesis and reject the alternate hypothesis, when in fact the null hypothesis is not true and the alternate is true; in other words, a false-negative error has been made. Beta is usually set at 0.20 in a one-sided arrangement in clinical research; the *power*, or ability of the study to detect a difference if one is present, is $1 - \beta$ and is therefore 0.80 if beta is 0.20 (Fig. 1).

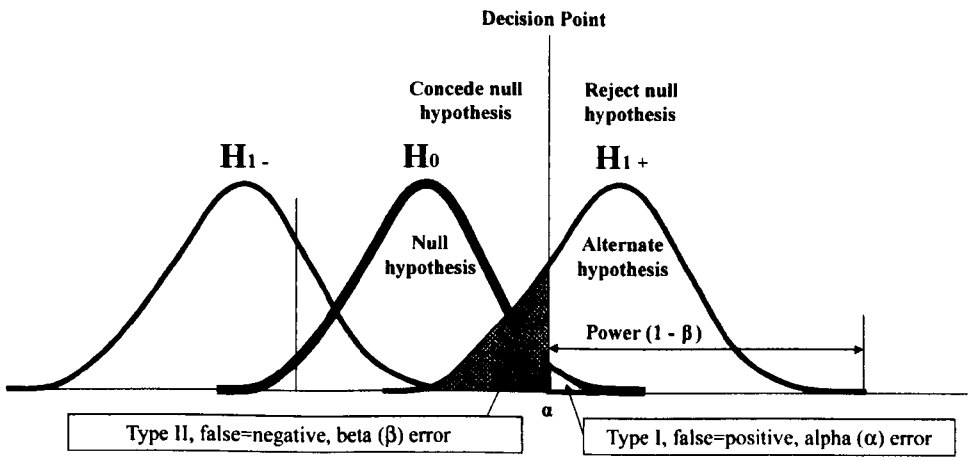


Fig. 1. Relationships of the probability curves of the null hypothesis and the alternate hypotheses, one for either of the two sides of the null curve. The null hypothesis curve displays an alpha (α) set at 5%, with 2.5% split between the two tails of a two-tailed arrangement. The decision point is at alpha. During sample size calculations, the beta (β) was set at 0.2 for a power of $1 - \beta$, or 0.80. The portion of the null hypothesis curve to the right of the decision point represents the type I error if the null hypothesis is rejected incorrectly. The portion of the alternate hypothesis curve to the left of the decision point depicts the type II error if the null hypothesis is conceded incorrectly.

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A power calculation after the study is done is not helpful in the reporting of the study; however, if necessary, it can help in reassessing published data to plan another study for definitive evidence.[9] Increasing the power of a study can be done by increasing the sample size; however, a generally preferable method is to decrease the measurement error by developing a more reliable and precise outcome measurement tool.[10]

Summary

Beta defines the false-negative error we are willing to accept if we accept the null hypothesis. *Power* defines the degree to which we are secure that a false-negative error was not made.

CONFIDENCE INTERVALS

A confidence interval (CI) defines the inner zone about the sample central index, mean, proportion, or ratio. It represents how much the observed sample central index might vary if multiple samples had been taken from the same general parent population.[1,5] It is estimated from the sample mean and the standard error of the mean (SEM) of the sample in continuous data, or from the proportion of interest (p) and the standard error (SE) of the proportion in proportional data. Confidence intervals also can be calculated for the difference between the means or proportions[5,11] in the comparison between the two samples.§

Confidence intervals state the degree of confidence we have that the population central index will fall within this range estimated from the sample data. For example, a 95% CI about a mean indicates that we are 95% confidant that the population mean will fall within the range described. Just as described in the sections above, alpha is set at the percentage of error we are willing to risk if we are wrong. For example, if alpha is set at 5% for the outer zone, 1 − alpha (95%) defines the inner zone. Because we are discussing the variability of the mean, rather than individual observation variability, the SD units for Z-score demarcations are SE units; thus, the sample mean ± 1.96 times the SEM defines the 95% CI about the sample mean.

Modern literature suggests that confidence intervals may be used to replace *P* values. In fact, both together give a better description of the data than either separately.[5] Important information is conveyed by confidence intervals, which show both the location and the width of the limits of confidence we have in the samples to represent the parent populations.[9,10]

§For single variable descriptions: CI = $X \pm Z\alpha$ (s/?N) for continuous data for large samples, and CI = $X \pm t\alpha_v$ (s/?N) for continuous data for small samples, where α is the same and v is the degrees of freedom; t is the t distribution normal curves for small sample sizes (usually less than 20); CI = $p \pm Z\alpha$ (?(pq/n)) for a binary proportion data. For bivariate analysis results, where the new mean or proportion is the difference between the compared sample means or proportions: CI = $(X_A - X_B) \pm t\alpha_v$(SED) for continuous variable comparisons, and CI = $p_A - p_B \pm Z\alpha$(SED) for the comparison of proportions.

Summary

A CI represents the degree of stability of the mean, or other central index; the wider the CI, the less stable the estimate and the less confident we are that it is a good estimate. The CI about the sample mean defines the range in which the population mean probably falls.

CONCLUSION

A clinical data set showing the contrast between two groups of subjects by three different measurement techniques yielding three different scales (dichotomous, ordinal, and continuous) was used as an example to demonstrate the application of three indexes of contrast (χ^2, Mann-Whitney *U*, and Student *t*-test). Each test generated a number called a *statistic*, which was associated with a probability score. The mathematics of calculation was performed by computer statistical programs; however, an explanation of what was being performed was included in the text and tables in this (part B) and the preceding article (part A).

The impact of sample size was emphasized. Sample size is an extremely important element to be considered in the design and in the analysis of a study. It is most efficient to define the smallest clinical difference that would be considered important during the sample size calculations; any smaller differences found, therefore, would be inconsequential. This allows consideration of both type I and type II errors in the project design to avoid a sample size that would be too large or too small.

The importance of measuring outcomes with as robust a method as possible was emphasized. Continuous data are more likely to show significant differences between groups than are dichotomous data. Ordinal data are the second best scale to use.

This tutorial on comparative statistics, which has been segmented into this article (part B) and the preceding one (part A), should be read in sequence and used together for ready reference on the topic. The challenge of writing such a tutorial was not the material but the need to explain statistical concepts understandably and efficiently to very bright, but also very busy, physicians. The more the clinician understands these few tests, the more efficiently and precisely he or she can critically read the literature and assimilate it into practice.

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